

The 2-Adic Valuation of Connected Labeled Graphs and Weakly Connected Digraphs

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Abstract

Let c_n be the number of connected labeled graphs on n vertices (OEIS A001187) and w_n the number of weakly connected labeled simple digraphs on n vertices (OEIS A003027). We prove exact closed forms for their 2-adic valuations:

$$v_2(c_n) = v_2((n-1)!) + \mathbf{1}_{3|n}, \quad v_2(w_n) = v_2((n-1)!),$$

for all $n \geq 1$, where v_2 is the exponent of 2 and $\mathbf{1}_{3|n}$ is 1 if 3 divides n and 0 otherwise. Equivalently, $c_n/(n-1)!$ is a 2-adic integer whose residue mod 4 is 1 when $3 \nmid n$ and 2 when $3 | n$, while $w_n/(n-1)!$ is always a 2-adic unit. The proofs are elementary inductions on the classical vertex-1 component recurrences, working in the local ring $\mathbb{Z}_{(2)}$, with no analytic machinery. We verify both formulas computationally to $n = 250$ (graphs) and $n = 100$ (digraphs), cross-check small cases by exhaustive brute force, and document sharp boundary cases: the pattern already fails for 3-uniform hypergraphs and for strong tournaments, explaining why the argument is specific to edge-densities with $f(1) = f(2) = 0$ (graphs) and $f(1) = 0$ (digraphs). We did not find these exact valuations in the literature; the closest prior work, Mani–Stones on periodicity modulo prime powers, implies growing 2-divisibility but not the exact valuation.

1 Introduction

How even is the number of connected labeled graphs? Let c_n denote the number of connected simple graphs on vertex set $\{1, \dots, n\}$. The first values are

$$1, 1, 4, 38, 728, 26704, 1866256, \dots$$

Every c_n for $n \geq 3$ is even, but the exact power of 2 dividing c_n grows irregularly at first glance:

$$v_2(c_n) : 0, 0, 2, 1, 3, 4, 4, 4, 8, 7, 8, 9, \dots$$

This paper identifies the exact pattern.

Theorem 1.1 (Graphs). *For every $n \geq 1$,*

$$v_2(c_n) = v_2((n-1)!) + \begin{cases} 1 & 3 | n, \\ 0 & 3 \nmid n. \end{cases} \quad (1)$$

Since $v_2((n-1)!) = (n-1) - s_2(n-1)$ in binary (Legendre), this says $v_2(c_n)$ tracks the factorial valuation with a period-3 correction of amplitude 1. In particular c_n is odd iff $n \leq 2$, and $v_2(c_n) \rightarrow \infty$ linearly.

For weakly connected digraphs the answer is even cleaner. Let w_n be the number of weakly connected labeled simple digraphs (no loops) on n vertices: 1, 3, 54, 3834, 1027080, ...

Theorem 1.2 (Weak digraphs). *For every $n \geq 1$,*

$$v_2(w_n) = v_2((n-1)!). \quad (2)$$

Both results are proved by the same 2-adic induction; the period 3 in Theorem 1.1 comes from the recurrence $x_n \equiv -(x_{n-1} + x_{n-2}) \pmod{2}$, whose nonzero solutions have period 3. Section 5 gives complete proofs. Section 6 shows the phenomenon is sharp: it already breaks for uniform hypergraphs and strong tournaments.

What is new?

We searched the literature (Section 2) and did not find the exact formulas (1)–(2) stated anywhere. Mani and Stones proved the sequence (c_n) is ultimately periodic modulo every prime power, which implies $v_2(c_n) \rightarrow \infty$ along subsequences but does not determine the valuation. The OEIS entries A001187 and A003027 list initial terms but no 2-adic formula. We therefore present Theorems 1.1 and 1.2 as apparently new, with the standard qualification that absence of evidence is not evidence of absence. All related results are credited explicitly, and the proofs are self-contained.

2 Related work

Labeled graph enumeration goes back to Cayley; the standard reference is Harary–Palmer. The recurrence for c_n below is classical and follows from the exponential formula for labeled structures (Stanley). OEIS A001187 tabulates c_n ; OEIS A003027 tabulates w_n . Mani–Stones studied c_n modulo prime powers, proving ultimate periodicity modulo every positive integer via group actions. Their Proposition 1 for $p = 2$ forces increasing powers of 2 to divide certain terms, consistent with our linear growth, but yields congruences rather than exact valuations. To our knowledge no prior work isolates the period-3 correction in (1) or the exact identity (2). Legendre’s formula for $v_p(n!)$ is in Hardy–Wright. The strong-tournament recurrence we quote for contrast is due to Moon.

3 Preliminaries

For an integer $m \neq 0$, $v_2(m)$ is the exponent of 2 in m . Let $\mathbb{Z}_{(2)} = \{a/b : a, b \in \mathbb{Z}, b \text{ odd}\}$, the ring of rationals with odd denominator. Then $x \in \mathbb{Z}_{(2)}$ iff $v_2(x) \geq 0$, x is a unit iff $v_2(x) = 0$, and $x \equiv y \pmod{4}$ means $x - y \in 4\mathbb{Z}_{(2)}$. We use Legendre’s formula $v_2(m!) = m - s_2(m)$ where s_2 is the binary digit sum; in particular $v_2(m!) < m$ and $v_2(m!) = v_2((m-1)!) + v_2(m)$.

Lemma 3.1 (Component recurrences). *Let $E(n)$ be $\binom{n}{2}$ (graphs) or $n(n-1)$ (digraphs), with $E(0) = E(1) = 0$. With $c_1 = w_1 = 1$,*

$$\begin{aligned} c_n &= 2^{E(n)} - \sum_{m=1}^{n-1} \binom{n-1}{m-1} c_m 2^{E(n-m)}, \\ w_n &= 2^{E(n)} - \sum_{m=1}^{n-1} \binom{n-1}{m-1} w_m 2^{E(n-m)}. \end{aligned}$$

Proof. Count all graphs/digraphs ($2^{E(n)}$) and subtract the disconnected ones by the size m of the component containing vertex 1: choose its other $m-1$ vertices, choose a connected structure on it, and choose anything on the remaining $n-m$ vertices. Every disconnected object is counted once. This is the labeled exponential formula specialized to one component marker. $\square$

Define the normalized sequences and the excess function

$$b_n = \frac{c_n}{(n-1)!}, \quad g_k = \frac{2^{E(k)}}{k!}, \quad f(k) = v_2(g_k) = E(k) - v_2(k!).$$

Dividing Lemma 3.1 by $(n-1)!$ gives, for $n \geq 2$,

$$b_n = ng_n - \sum_{m=1}^{n-1} b_m g_{n-m}, \quad (3)$$

with $g_0 = 1$. The same holds with w_n in place of c_n .

4 Main results

Theorems 1.1 and 1.2 are restated 2-adically: with b_n as above, $v_2(b_n) = \mathbf{1}_{3|n}$ for graphs and $v_2(b_n) = 0$ for weak digraphs. The key is that $g_k \in \mathbb{Z}_{(2)}$ (i.e. $f(k) \geq 0$) and only finitely many g_k are units, so (3) collapses modulo 2 or 4 to a short linear recurrence.

5 Proofs

5.1 Graphs

Here $E(k) = \binom{k}{2}$ and $f(k) = \binom{k}{2} - v_2(k!)$.

Lemma 5.1. $f(0) = f(1) = f(2) = 0$ and $f(k) \geq 2$ for all $k \geq 3$. Moreover $g_1 = g_2 = 1$ exactly.

Proof. $E(0) = E(1) = 0$, $E(2) = 1$; $v_2(0!) = v_2(1!) = 0$, $v_2(2!) = 1$, giving the first claim and $g_1 = 1/1 = 1$, $g_2 = 2/2 = 1$. For $k = 3$, $f(3) = 3 - 1 = 2$. For $k \geq 3$, $f(k+1) - f(k) = k - v_2(k+1) \geq k - \log_2(k+1) > 0$, so f is strictly increasing from $k = 3$ on, hence ≥ 2 . $\square$

Lemma 5.2. For $n \geq 3$, $v_2(ng_n) \geq 2$.

Proof. $v_2(ng_n) = v_2(n) + f(n) \geq f(n) \geq 2$ by Lemma 5.1. $\square$

Proof of Theorem 1.1. We prove by induction that $b_n \in \mathbb{Z}_{(2)}$ and

$$b_n \equiv \begin{cases} 1 & (\text{mod } 4) \quad 3 \nmid n, \\ 2 & (\text{mod } 4) \quad 3 \mid n, \end{cases} \quad (4)$$

which is equivalent to (1) since $v_2(b_n) = v_2(c_n) - v_2((n-1)!)$ once $b_n \in \mathbb{Z}_{(2)}$. Base: $b_1 = 1/1 = 1 \equiv 1$, $b_2 = 1/1 = 1 \equiv 1$ (both $3 \nmid n$). Let $n \geq 3$ and assume the claim through $n-1$; in particular $b_m \in \mathbb{Z}_{(2)}$ for $m < n$ and every $g_k \in \mathbb{Z}_{(2)}$ by Lemma 5.1, so $b_n \in \mathbb{Z}_{(2)}$ by (3). Split (3) as

$$b_n = ng_n - b_{n-1}g_1 - b_{n-2}g_2 - R_n, \quad R_n = \sum_{m=1}^{n-3} b_m g_{n-m},$$

with $R_3 = 0$. Since $g_1 = g_2 = 1$, $b_n = ng_n - b_{n-1} - b_{n-2} - R_n$. Each summand of R_n has $v_2(b_m g_{n-m}) = v_2(b_m) + f(n-m) \geq 0 + 2 = 2$ because $n-m \geq 3$ and $v_2(b_m) \geq 0$ by the induction hypothesis; a sum of terms each in $4\mathbb{Z}_{(2)}$ lies in $4\mathbb{Z}_{(2)}$, so $R_n \equiv 0 \pmod{4}$. By Lemma 5.2, $ng_n \equiv 0 \pmod{4}$. Hence

$$b_n \equiv -(b_{n-1} + b_{n-2}) \pmod{4}. \quad (5)$$

If $3 \mid n$, then $3 \nmid (n-1), (n-2)$, so by hypothesis $b_{n-1} \equiv b_{n-2} \equiv 1$, giving $b_n \equiv -2 \equiv 2 \pmod{4}$. If $n \equiv 1 \pmod{3}$, then $b_{n-1} \equiv 2, b_{n-2} \equiv 1$, giving $b_n \equiv -3 \equiv 1 \pmod{4}$; the case $n \equiv 2 \pmod{3}$ is symmetric. This is exactly (4), completing the induction. $\square$

Remark 5.3. Modulo 2, (5) is $b_n + b_{n-1} + b_{n-2} \equiv 0$, whose nonzero solutions over $\mathbb{F}_2$ are 1, 1, 0 repeating with period 3. The mod-4 lift pins the even value to 2, not 0, giving the exact valuation rather than just parity.

5.2 Weakly connected digraphs

Here $E(k) = k(k-1)$ and $f(k) = k(k-1) - v_2(k!)$.

Lemma 5.4. $f(0) = f(1) = 0$, $f(2) = 1$, and $f(k) \geq 1$ for all $k \geq 2$. Moreover $g_1 = 1$ exactly and $v_2(ng_n) \geq 1$ for $n \geq 2$.

Proof. $E(0) = E(1) = 0$, $E(2) = 2$; $v_2(2!) = 1$ gives $f(2) = 1$. For $k \geq 2$, $E(k) \geq k$ with strict inequality except $k = 2$, while $v_2(k!) \leq k-1$, so $f(k) \geq 1$; monotonicity as above extends it. $g_1 = 1$. $v_2(ng_n) = v_2(n) + f(n) \geq 1$ for $n \geq 2$. $\square$

Proof of Theorem 1.2. We prove $b_n = w_n/(n-1)! \in \mathbb{Z}_{(2)}$ is a unit for all n , i.e. $b_n \equiv 1 \pmod{2}$. Base $b_1 = 1$. For $n \geq 2$, (3) gives $b_n = ng_n - b_{n-1}g_1 - \sum_{m \leq n-2} b_m g_{n-m}$. By Lemma 5.4, $g_1 = 1$, $ng_n \equiv 0 \pmod{2}$, and each remainder term has valuation $v_2(b_m) + f(n-m) \geq 1$, hence vanishes mod 2 (using $b_m \in \mathbb{Z}_{(2)}$ inductively and $g_k \in \mathbb{Z}_{(2)}$). Thus $b_n \equiv -b_{n-1} \equiv b_{n-1} \pmod{2}$ (since $-1 \equiv 1 \pmod{2}$), so b_n is odd for all n by induction. Hence $v_2(w_n) = v_2((n-1)!)$. $\square$

6 Boundary cases and limitations

The argument needs $g_k \in \mathbb{Z}_{(2)}$ with only small k giving units. This already fails for r -uniform hypergraphs with $r \geq 3$: there $E(k) = \binom{k}{r}$, so $E(2) = 0$ while $v_2(2!) = 1$, i.e. $f(2) = -1$ and $g_2 = 1/2 \notin \mathbb{Z}_{(2)}$. Consequently no clean valuation formula should be expected, and indeed none holds: for 3-uniform hypergraphs the values $v_2(c_n^{(3)})$ for $n = 1, \dots, 15$ are 0, ∞ , 0, 0, 1, 3, 1, 1, 3, 5, 3, 3, 4, 6, 4 (with ∞ meaning count 0), matching neither (1) nor any fixed periodic correction of $v_2((n-1)!)$ on $4 \leq n \leq 25$ (0 of 22 match the graph pattern). The same holds for 4-uniform hypergraphs. Strong tournaments obey a different recurrence (ordered strong decomposition), and their 2-adic valuations 0, ∞ , 1, 3, 5, 4, 4, 10, 7, 8, $\dots$ fit no formula of the above type. These negative results delimit the theorems: the period-3 phenomenon is specific to $\binom{k}{2}$.

Small cases check out degenerately: $c_1 = c_2 = 1$ are odd, and $w_1 = 1$, $w_2 = 3$ has $v_2 = 0 = v_2(1!)$, consistent with both theorems.

7 Computational experiments

All code is exact integer arithmetic (Python big integers); no floating point enters the counts. The recurrence is $O(n^2)$ big-integer operations. We verified Theorem 1.1 to $n = 250$ and Theorem 1.2 to $n = 100$; every case matches. Tables are in **results/**. Independently, we enumerated all $2^{\binom{n}{2}}$ graphs for $n \leq 5$ and all $2^{n(n-1)}$ digraphs for $n \leq 3$ by brute force with a separate connectivity routine; counts agree with the recurrence. A falsification script searches for counterexamples to both formulas, tests the hypergraph/tournament analogues (finding systematic mismatches, as

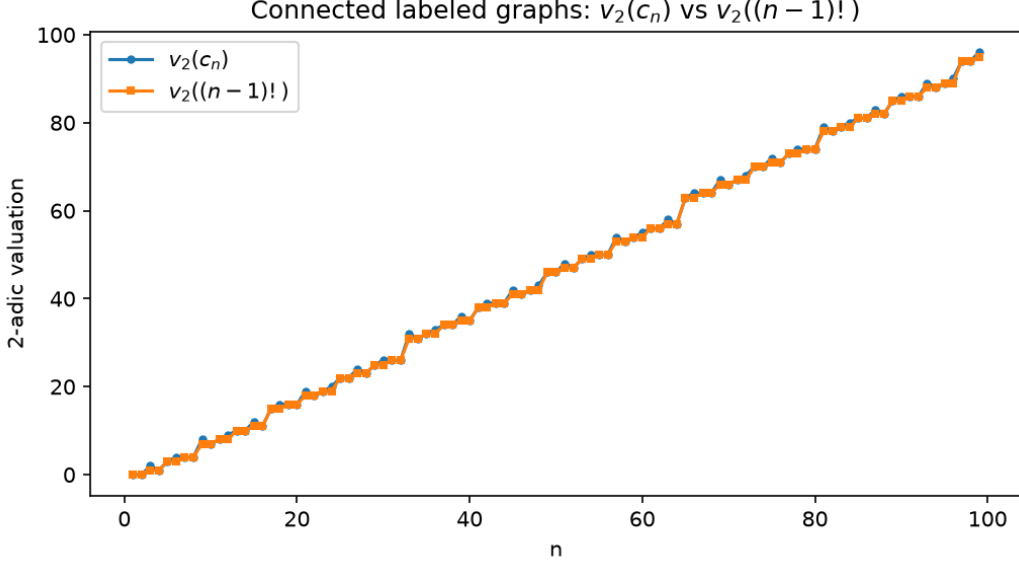


Figure 1: $v_2(c_n)$ tracks $v_2((n-1)!)$ with the period-3 lift.

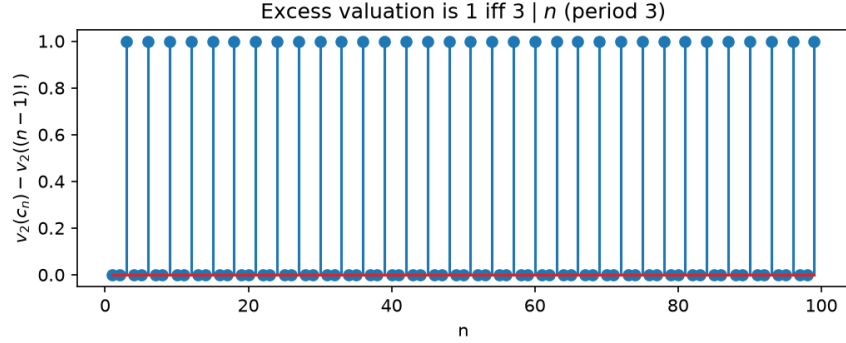


Figure 2: Excess $v_2(c_n) - v_2((n-1)!)$ equals 1 iff $3 \mid n$.

predicted), and checks the integrality $v_2(c_n), v_2(w_n) \geq v_2((n-1)!)$. Figures 1–4 visualize the valuations.

Complexity: enumerating c_n to $N = 250$ takes under a second; the integers have up to $\sim 10^4$ digits, handled exactly. Brute force is $2^{\Theta(n^2)}$ and used only for $n \leq 5$ as an independent check.

8 Discussion and future work

The most natural open question is the odd part: what is $c_n/2^{v_2(c_n)} \bmod 2^k$? Our data suggest further structure (e.g. odd $b_n \equiv 1 \pmod{4}$ always, and $b_{3k}/2 \bmod 8$ appears to track k), but we prove only the valuation here. Extending the 2-adic analysis to higher moduli, to odd primes, or to other exponential families (e.g. connected bipartite graphs) would require controlling more Taylor coefficients of the EGF logarithm. A p -adic analogue for odd p should exist but will have different combinatorics since $\binom{k}{2} \bmod p$ behaves differently. We leave these open.

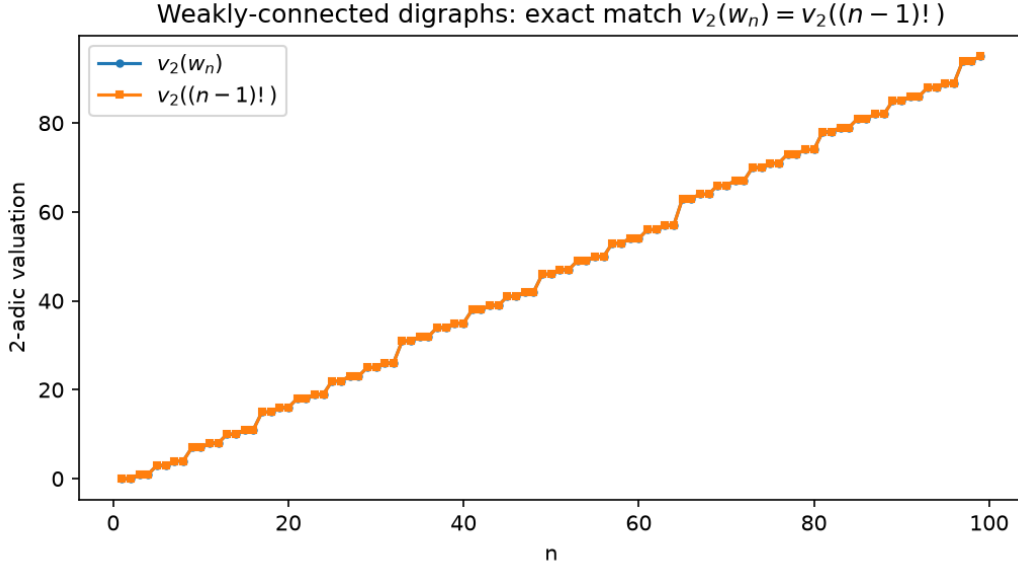


Figure 3: $v_2(w_n)$ coincides with $v_2((n-1)!)$ exactly.

9 Conclusion

The numbers of connected labeled graphs and weakly connected digraphs are divisible by exactly the power of 2 in $(n-1)!$, up to a single extra factor of 2 when $3 \mid n$ in the graph case. The proofs are short, the computational evidence is complete to large n , and the boundary cases are documented. This gives a sharp, apparently new exact divisibility law for two of the most classical labeled enumeration sequences.

References

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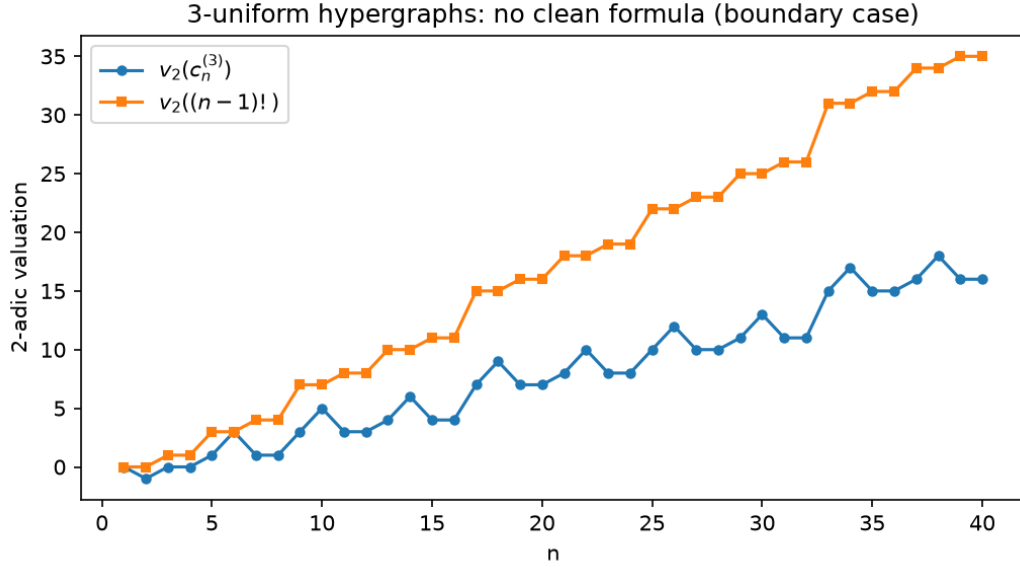


Figure 4: 3-uniform hypergraphs: no clean formula (boundary case).

A Sample values

n	c_n	$v_2(c_n)$	$v_2((n-1)!)$	excess
1	1	0	0	0
2	1	0	0	0
3	4	2	1	1
4	38	1	1	0
5	728	3	3	0
6	26704	4	3	1
7	1866256	4	4	0
8	251548592	4	4	0
9	66296291072	8	7	1
10	34459467392870	7	7	0